

FIRST YEAR PROJECT

GROUP: XERUS

Edvinas Gross, edgr@itu.dk
Eduardo Goncalves Ramos de Almeida, edal@itu.dk
Sameer Malkawi, smal@itu.dk
Damian Wojciech Wardzinski, damw@itu.dk
Maksymilian Rafal Wozniak, makw@itu.dk

<https://github.com/edgrobbit/2026-PDS-XERUS>

PROJECTS IN DATA SCIENCE

1 Introduction

Skin cancer is an important public health problem, and early detection can improve the chances of successful treatment. Skin cancers are among the most commonly diagnosed cancer groups worldwide, with more than 1.5 million new cases estimated in 2022. Melanoma is less frequent than non-melanoma skin cancers, but it is more dangerous because it has a higher risk of spreading if not detected early. American Cancer Society (2024)

Dermatologists assess suspicious lesions through visual inspection, dermoscopy, and clinical information about the patient. One common diagnostic guideline is the ABC rule, which considers Asymmetry, Border irregularity, Colour variation. These visual characteristics are useful because malignant lesions often present irregular shapes, uneven borders, and abnormal colour patterns. Nachbar et al. (1994)

In this project, we aim to build a computer-aided model for classifying skin lesions as cancerous or non-cancerous using clinical images from the PAD-UFES-20 dataset. While previous approaches often focus on lesion-specific features such as border shape, asymmetry, compactness, and colour variation inside the lesion, our group investigates whether adding information about the surrounding skin colour / skin tone can improve the model.

The research question guiding this project is:

Can skin colour detection improve the performance of a skin lesion classification model?

This question is relevant because skin colour may affect how lesions appear in images.

2 Related Work

Skin tone has become an increasingly important topic in dermatology AI. Several studies have shown that AI models may perform worse on darker skin tones, partly because many dermatology datasets contain fewer examples from patients with darker skin. Daneshjou et al. found that dermatology AI models showed limitations on a diverse clinical image dataset, especially for darker skin tones and uncommon diseases. Daneshjou et al. (2022)

Recent research argues that dermatology datasets should include better skin tone labels. Weir et al. note that skin tone affects AI performance in dermatology and that better skin tone labeling may improve model generalizability for detecting dermatologic malignancies. They also emphasize that Fitzpatrick Skin Type is not exactly the same as skin tone, since it was originally designed to describe skin reaction to sun exposure rather than objective colour. Benčević et al. (2024)

For our project, this literature supports the idea that skin colour detection is not only a technical feature extraction problem, but also a fairness and generalization issue. If skin colour affects image contrast and model predictions, then adding a skin colour feature may improve classification performance.

3 Data

This project uses the PAD-UFES-20 dataset, the same dataset used in the previous project. PAD-UFES-20 is a public skin lesion dataset collected by the Dermatological and Surgical Assistance Program at the Federal University of Espírito Santo in Brazil. The dataset contains clinical images collected using smartphones. Pacheco et al. (2020)

The dataset contains 2,298 images from 1,373 patients. Each sample includes a clinical image and metadata such as patient age, lesion location, Fitzpatrick skin type, and lesion diameter. The images are stored in PNG format and have different sizes because they were collected using different smartphone devices.

We have also been provided with 2,151 masks. In relation to data exploration, we found 195 images without masks, 48 masks without images, and 2 images with incorrect name structures. After processing, we considered only the ones with masks and not mismatched, leaving 2,100 unique data points. The malignant and benign lesions are distributed evenly, with the malignant count being at 1,052 images, and benign at 1,048.

The original lesion diagnoses include:

- Basal Cell Carcinoma (BCC)
- Squamous Cell Carcinoma (SCC)
- Malignant Melanoma (MEL)
- Actinic Keratosis (ACK)
- Melanocytic Nevus (NEV)
- Seborrheic Keratosis (SEK)

We group the diagnoses into two broader classes:

- **Cancerous:** BCC, SCC, MEL
- **Non-cancerous:** ACK, NEV, SEK

This binary classification is useful because some diagnostic categories, especially melanoma, have relatively few observations. Grouping the labels helps reduce the effect of class imbalance and makes the task more focused on distinguishing potentially malignant lesions from non-cancerous ones.

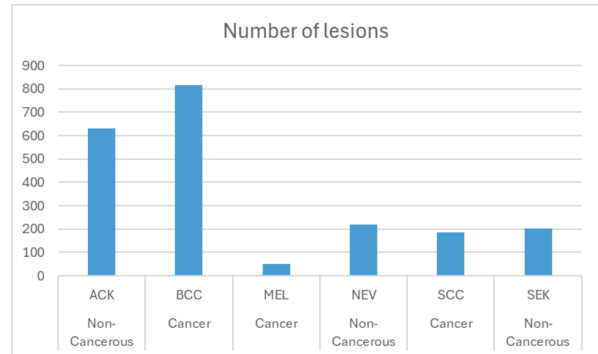


Figure 1: Number of lesions by category

4 Baseline Classification Model

4.1 Feature Extraction

4.1.1 Asymmetry

We evaluated the asymmetry score of the binary mask by XORing mirrored halves and computing the non-overlapping pixels at multiple orientations (manually changeable by altering the n value), to reduce the sensitivity to the lesion's orientation in the image. The output is later normalised to 0–1, where higher values indicate greater shape irregularity.

The mean of the asymmetry score of each mask subsequently has been saved in a CSV file that will be one of the baseline parameters for the machine learning models.

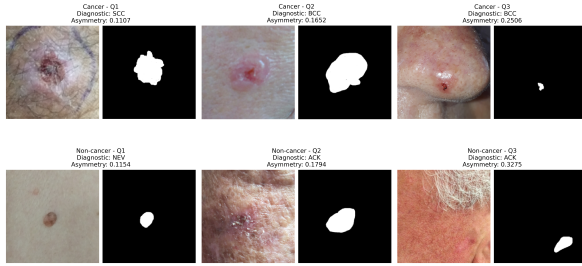


Figure 2: Asymmetry feature extraction examples

4.1.2 Borders

For borders, we would have trouble with masks which have multiple lesions. The treatment we chose for them was to eliminate all minor lesions and only use the biggest lesion for the sake of border variable calculations, as showcased in the image:

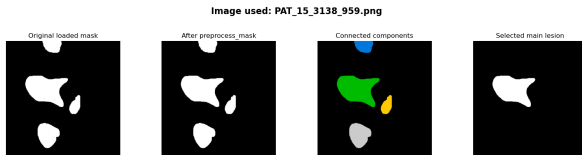


Figure 3: Mask preprocessing to isolate the main lesion

Compactness was used as one of the border-related features extracted from the lesion masks. It measures how close the shape of a lesion is to a perfect circle by comparing its perimeter and area. Compactness was calculated using:

$$\text{Compactness} = \frac{\text{Perimeter}^2}{4\pi \times \text{Area}}$$

A perfectly circular lesion has a compactness value close to 1. As the lesion becomes more irregular, elongated, or has a more complex border, the compactness value increases. Therefore, higher compactness values may indicate that the lesion has a less regular shape.

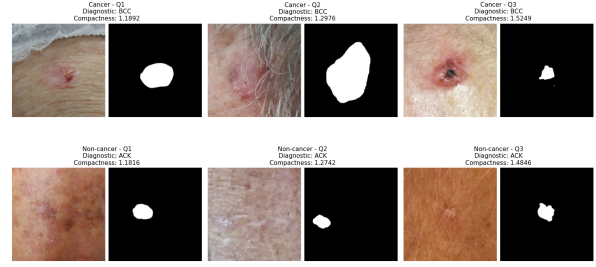


Figure 4: Compactness feature extraction examples

This is useful in skin lesion analysis because malignant lesions can often present more irregular borders compared to benign lesions. By including compactness as a feature, the model can use information about the shape and border complexity of the lesion, rather than relying only on size-based measurements such as area or lesion percentage.

In our code, compactness is calculated from the binary lesion mask using the lesion area and perimeter. This means that the quality of the mask is important, because noisy or poorly segmented borders can affect the perimeter and therefore change the compactness value. For this reason, the masks were preprocessed before feature extraction by removing small objects and filling small holes.

Lesion percentage was included as a relative size feature. It was calculated by dividing the number of white pixels in the segmentation mask, which represent the lesion area, by the total number of pixels in the image. This gives the proportion of the image occupied by the lesion. Unlike the raw area, lesion percentage makes the feature easier to compare across images, especially when images may have different sizes or resolutions.

4.1.3 Pen Mark Mask Creation

Pen marks on skin lesion images introduce visual artifacts that can distort skin tone and lesion boundaries. To address this, the lesion region is first temporarily occluded by replacing it with the median RGB value of the surrounding skin pixels, preventing the discoloured lesion from interfering with detection.

Skin tone is then estimated by computing the weighted grayscale mean of skin pixels using the standard luminance formula:

$$g = 0.114B + 0.587G + 0.299R$$

classifying images below 120 as dark skin and above as light skin. This determines which of two adaptive detection strategies is applied.

For dark skin, detection combines an HSV-based hue mask targeting blue tones and a darkness mask targeting black pen marks. For light skin, a statistically robust approach is used, computing the median and median absolute deviation (MAD) of both pixel brightness and the blue-red channel shift ($B - R$), flagging pixels that deviate significantly from the skin baseline in either measure.

The resulting mask is refined through morphological opening and closing to remove noise and fill gaps. Directional dilation is then applied along horizontal, vertical,

and diagonal axes, using a voting scheme that retains only pixels agreed upon by at least two directions, ensuring coverage of pen strokes without excessive mask bloating. Finally, connected components smaller than 80 pixels are discarded, and the confirmed pen regions are inpainted using the Telea algorithm, which propagates surrounding skin texture inward to produce a clean reconstruction.

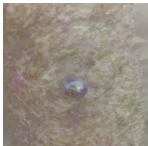
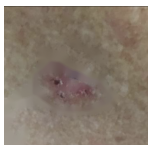
Sample & Score	Before	After
Case 1 (Score: 1)		
Case 2 (Score: 1)		
Case 3 (Score: 1)		

Table 1: Pen mark removal results

4.1.4 Colors

While lesion-specific features such as asymmetry, border irregularity, and compactness describe the lesion itself, the overall colour distribution of the image may also carry diagnostic value. Skin colour affects image contrast, and a lesion that stands out clearly against light skin may appear differently against darker skin tones, potentially influencing both human assessment and model predictions. As discussed in the Related Work section, AI models in dermatology have been shown to perform unevenly across skin tones, partly because training datasets underrepresent patients with darker skin. By explicitly encoding colour as a feature, we aim to give the model awareness of this variation rather than leaving it as an uncontrolled source of noise.

Colour features are extracted from the full clinical image using two complementary methods: superpixel-based variance analysis. Images are resized to a maximum dimension of 512 pixels before processing to ensure consistent scale across the dataset, since the PAD-UFES-20 images were captured on different smartphone devices with varying resolutions.

All colour analysis is performed in HSV (Hue, Saturation, Value) space, with RGB retained for variance calculations. HSV is better suited to perceptual colour analysis than raw RGB because it separates chromatic information (hue and saturation) from brightness (value), making it less sensitive to lighting variation across images.

Each image is segmented into approximately 50 superpixels using the SLIC algorithm. SLIC groups spatially adjacent pixels with similar colour into compact regions, producing a more meaningful unit of analysis

than individual pixels. For each superpixel, the mean RGB and HSV values are computed. The variance across all superpixel means is then calculated for each channel separately, yielding six scalar features per image: variance of R, G, B, H, S, and V. These variance values capture how heterogeneous the colour distribution is across the image — higher variance in hue or saturation may indicate a more complex or irregular lesion presentation. Achanta et al. (2012)

Hue variance is computed using circular variance to correctly handle the wrap-around nature of the hue channel. This leads to six variance features, which are subsequently merged with the lesion geometry features for model training.

To capture meaningful colour information from each dermoscopic image, three sets of statistical features were extracted and included as direct inputs to the model: RGB channel variance, HSV channel variance, and the mean Individual Typology Angle (ITA). For each image, the per-channel variance was computed across both the RGB and HSV colour spaces, yielding six features in total. RGB variance captures the spread of raw colour intensity across the red, green, and blue channels, while HSV variance separates hue (colour tone), saturation (colour purity), and value (brightness), which can be more robust to lighting variation than raw RGB values. These features collectively encode information about colour heterogeneity within the lesion.

The ITA mean was additionally computed per image as a quantitative measure of skin tone as:

$$\text{ITA} = \arctan\left(\frac{L - 50}{b^*}\right) \times \frac{180}{\pi}$$

L = lightness, b=Blue-Yellow axis

where higher values correspond to lighter skin tones and lower values to darker ones. Images were then stratified into six skin tone groups following the standard Fitzpatrick-style ITA bands. This stratification enabled a post-hoc analysis of whether model performance varies across different skin tones, which is particularly relevant in dermoscopy where darker skin tones are historically underrepresented in training datasets, potentially leading to biased model behaviour. Notably, no colour normalisation or balancing was applied to the images themselves; the ITA grouping was purely analytical, intended to surface any disparities in model performance rather than to alter the data distribution.

Skin Classification	Individual Typology Angle, ITA°
Very Light	ITA° > 55°
Light	41° < ITA° ≤ 55°
Intermediate	28° < ITA° ≤ 41°
Tan	10° < ITA° ≤ 28°
Brown	−30° < ITA° ≤ 10°
Dark	−50° < ITA° ≤ −30°
Very Dark	ITA° ≤ −50°

Table 2: Skin classification based on ITA°

4.1.5 Hair Removal Feature

When taking close-up pictures of skin lesions, body hair often gets in the way. This is an issue because it can confuse our classification models, making them think the lines from the hair are part of the actual skin condition.

To fix this, we created an automated step to clean up the pictures before feeding them into the model. First, we estimate how much hair is actually in the picture so we don't accidentally ruin a clean image. We do this by extracting the green colors from the photo, which happens to provide the best natural contrast between dark hair and lighter skin. To make the hair stand out even more, we boost the contrast using a technique called CLAHE (Contrast Limited Adaptive Histogram Equalization). Instead of applying a blanket contrast adjustment to the whole image—which can wash out the details—CLAHE smartly adjusts the contrast in small, localized sections. We then use a few edge-detecting tools to spot both thick and thin hairs, combining them to create a black-and-white map of where the hair is. We also make sure the computer ignores the dark, circular borders that these cameras naturally leave around the edges of the photo.

If the image has less than 0.5% hair coverage, we just leave it alone to avoid creating unnecessary digital smudges. For the pictures that do need cleaning, we use image filters called BlackHat and TopHat transforms. BlackHat is specifically designed to isolate dark, narrow shapes against a lighter background (perfect for separating dark hair from skin), while TopHat does the reverse. We run these filters using an adjustable kernel (a specific shape the computer uses to scan the image) that changes size depending on whether the hair is sparse or dense. We also use adaptive filtering to account for bad or uneven lighting across the skin. To actually erase the hair and fix the image, we used the Telea algorithm. Telea (2004)

What it does is once the hair is removed, it leaves empty pixels behind. The Telea algorithm looks at the healthy skin immediately surrounding those empty spots and seamlessly pulls those colors inward to fill the gap. We chose this specific method because it is highly effective at keeping the skin looking natural. Finally, we ran this automated cleanup across our entire dataset. We saved the cleaned images under their original file names to keep our data organized, and we manually compared the before-and-after pictures to ensure the skin lesions themselves weren't accidentally altered.

For scores we used in the tables we used agreed on annotations from the preliminary assignment, where it measures hair density (0 - no hair, 3 - very dense).

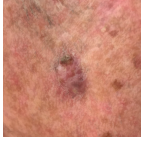
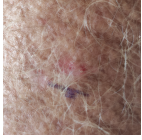
Sample & Score	Before	After
Case 1 (Score: 2 or 1)		
Case 2 (Score: 3 or 2)		
Case 3 (Score: 3)		

Table 3: Hair removal results

5 Models

Our database splits training and sample data into the 80:20 proportion, with both sets of data also following the same splits as our database, balanced almost 50/50 in non-cancers and cancer cases. Since our database is balanced, we proceeded with evaluation. On all our models we made 3 different scenarios:

1. Only using shape features (asymmetry + border)
2. Incorporating color features (RGB variance and HSV variance)
3. Incorporating skin color (ITA mean)

5.1 KNN

The first model test we performed is the K-Nearest neighbour, which operates on the principle that similar data points are located close to each other in the feature space, meaning predictions for a new data point are derived from the k closest training examples. Out of 3 models KNN has performed the weakest, since it performs best on clean, low-variance data where distance meaningfully reflects similarity, and PAD-UFES-20 introduced a lot of RGB/HSV variance due to the unorganized data collection approach and outside disturbances such as hair and penmarks.

Clinical data was missing roughly 35% of Fitzpatrick values. We tried to re-evaluate the skin type on the cleaned up images, however the predicted FST distribution skews heavily toward darker skin tones, going from 5.2 % of the patients being in the 3-6 range (Olive skinned to dark brown), to 53.9% patients being within the same range according to ITA predictions. This skew is a direct consequence of the inconsistent imaging conditions in PAD-UFES-20, images taken with inconsistent photography devices and under distances and lighting that varies, causes the skin to appear more dark than it actually is, causing ITA values to systematically underestimate skin lightness. This means that the feature fed into Model C does not reliably represent true skin tone, but rather a combination of skin tone and lighting noise. This small improvement from Model C

compared to Model B should be interpreted with caution, as skin color information was already distorted before reaching the model.

FST	Tone	FST Pred (n)	Pred (%)	FST Ann (n)	Ann (%)
1	Very pale	229	10.9%	143	10.9%
2	Fair	339	16.1%	839	63.9%
3	Medium	398	18.9%	368	28.0%
4	Olive	568	27.0%	58	4.4%
5	Brown	517	24.6%	9	0.7%
6	Dark brown	49	2.3%	1	0.1%
Total		2100	100%	1418 (804 miss)	100%

Table 4: Fitzpatrick Skin Type Prediction vs Annotation

Our KNN model cross validates on the training set and chooses the maximizing k-value that maximizes AUC on the validation folds. AUC was preferred over accuracy as it evaluates the model’s ability to distinguish between classes across all decision thresholds, making it a more informative metric for comparing model variants.

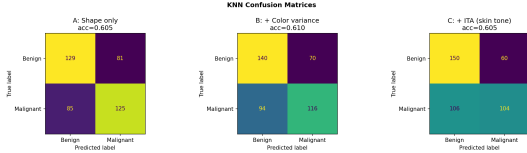


Figure 5: KNN Confusion matrices across models

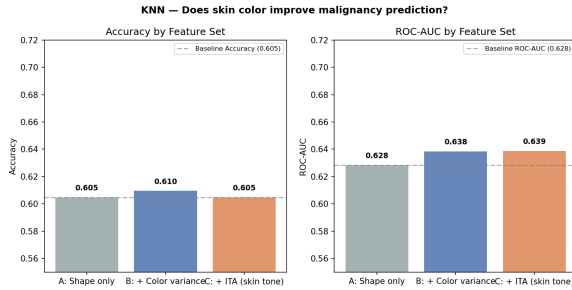


Figure 6: KNN comparison across models

• Model A: Shape only

- Features used: asymmetry score, compactness, and lesion percentage.
- Best k: 19.
- Test accuracy: 0.605.
- Test AUC: 0.628.
- This was the weakest KNN model among the three tested versions.

• Model B: Shape + color variance

- Features used: shape features plus RGB and HSV color variance features.
- Best k: 19.

- Test accuracy: 0.610.
- Test AUC: 0.638.
- This model improved over the shape-only model in both accuracy and AUC.
- Adding color variance features produced a slight increase in performance at recognizing malignant lesions.

• Model C: Shape + color variance + ITA

- Features used: all previous features plus ITA mean, representing skin tone.
- Best k: 20.
- Test accuracy: 0.605.
- Test AUC: 0.639.
- Adding ITA slightly improved AUC compared with Model B, but accuracy decreased.
- The model became slightly more sensitive to malignant lesions, though at the cost of increased false positives.

The KNN performance was less than desirable. The confusion matrix shows that there were too many FN for this model to be actually viable in a medical environment, though it did have significantly more FP. Adding skin tone as a feature did not meaningfully improve the model’s performance, though it would be interesting to see the change in performance on data collected under more controlled and consistent imaging conditions.

5.2 Logistic Regression

Logistic regression is used to predict two possible outcomes, in our project whether a lesion is cancerous or non-cancerous. Instead of giving only a final answer, it first calculates the probability of an image belonging to one class. Then, using a threshold such as 0.5, it classifies the image into one of the two groups. A logistic regression also allows us to see which features are more related to the probability of cancer. After testing the three models, the results were:

The logistic regression models were tested to answer the research question: *Does skin color influence malignancy prediction?* Three models were compared: a shape-only model, a model including color variance features, and a model including ITA as a skin tone feature.

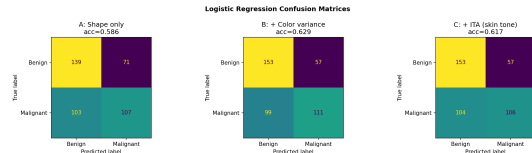


Figure 7: Logistic regression Confusion matrices across models

• Model A: Shape only

- Features used: asymmetry score, compactness, and lesion percentage..

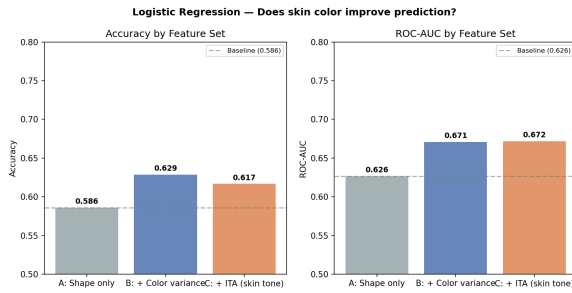


Figure 8: Logistic regression comparison across models

- Test accuracy: 0.586.
- Test AUC: 0.626.
- This was the weakest logistic regression model among the three tested versions.
- The model performed better at identifying benign lesions, with a benign recall of 0.66.
- However, the malignant recall was only 0.51, meaning that the model correctly detected only 51% of malignant lesions.
- This suggests that using only shape-related features was not enough to strongly separate benign and malignant lesions.

• Model B: Shape + color variance

- Features used: shape features plus RGB and HSV color variance features.
- Test accuracy: 0.629.
- Test AUC: 0.671.
- This model improved over the shape-only model in both accuracy and AUC.
- It had the highest test accuracy among the three logistic regression models.
- The benign recall was 0.73, meaning that the model correctly identified 73% of benign lesions.
- The malignant recall was 0.53, meaning that the model still missed many malignant cases.
- Overall, adding color variance features improved the model's ability to distinguish between benign and malignant lesions.

• Model C: Shape + color variance + ITA

- Features used: all previous features plus ITA mean, representing skin tone.
- Test accuracy: 0.617.
- Test AUC: 0.672.
- This model had the highest test AUC, but only by a very small margin compared with Model B.
- Its accuracy was slightly lower than Model B, decreasing from 0.629 to 0.617.
- The benign recall remained high at 0.73, but the malignant recall decreased to 0.50.

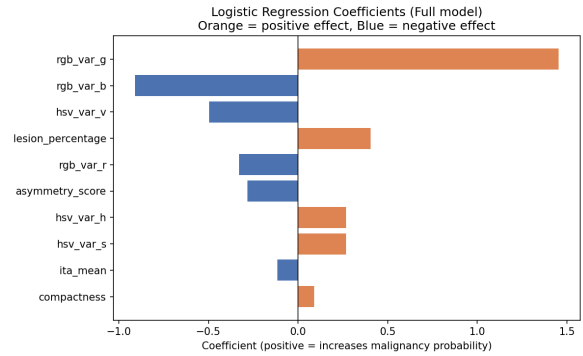


Figure 9: Logistic regression coefficients

- This means that adding ITA did not improve the detection of malignant lesions.

In terms of coefficients, the strongest positive predictor in both Model B and Model C was RGB green variance, a color feature. In Model B, RGB green variance had a coefficient of 1.4071 and an odds ratio of 4.0842. In Model C, it had a coefficient of 1.4548 and an odds ratio of 4.2836. This suggests that higher RGB green variance was associated with higher predicted odds of malignancy in the logistic regression model.

Lesion percentage remained a relevant positive predictor across all three models. This suggests that the relative size of the lesion in the image contributed to malignancy prediction.

The ITA mean feature, which represents skin tone, had a relatively small effect in Model C, which indicates that ITA had only a limited contribution to the logistic regression model.

Overall, the logistic regression results suggest that adding color variance features improved the model compared with using shape features only. Model B improved the test AUC from 0.626 to 0.671 and increased accuracy from 0.586 to 0.629. Model C achieved a slightly higher AUC of 0.672, but this improvement over Model B was very small, while accuracy decreased. Therefore, these results suggest that lesion color variation may help malignancy prediction, but there is no strong evidence that ITA, used as a skin tone feature, meaningfully managed to improve the model.

5.3 Decision Tree

A decision tree is used to make predictions by splitting the data into smaller groups based on the values of the features. The model works like a sequence of questions, where each answer leads to another split until a final classification is made. However, decision trees can also overfit the data if they become too deep, so it is important to control their depth or use pruning.

• Model A: Shape only

- Features used: asymmetry score, compactness, and lesion percentage.
- Best tree depth: 4.
- Test accuracy: 0.600.

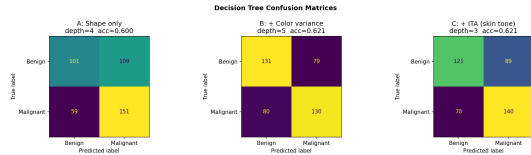


Figure 10: Decision Tree Confusion matrices across models

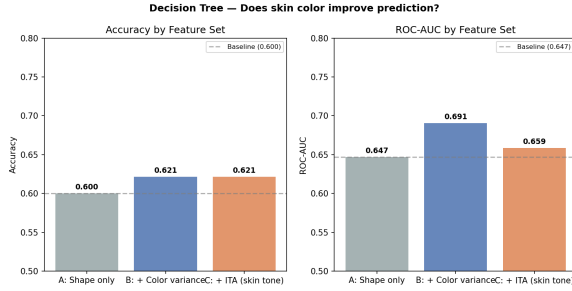


Figure 11: Decision Tree comparison across models

- Test AUC: 0.647.
- The malignant recall was 0.72, meaning that the model correctly identified 72% of malignant lesions.
- However, the benign recall was only 0.48, meaning that many benign lesions were incorrectly classified as malignant.

• Model B: Shape + color variance

- Features used: shape features plus RGB and HSV color variance features.
- Best tree depth: 5.
- Test accuracy: 0.621.
- Test AUC: 0.691.
- The recall for both benign and malignant lesions was 0.62, showing a balanced performance between the two classes.
- These results suggest that adding color variance features improved the model compared with using shape features only.

• Model C: Shape + color variance + ITA

- Features used: all previous features plus ITA mean, representing skin tone.
- Best tree depth: 3.
- Test accuracy: 0.621.
- Test AUC: 0.659.
- This model had the same accuracy as Model B, but a lower AUC.
- The malignant recall was 0.67, meaning it detected more malignant cases than Model B. However, the benign recall was lower, at 0.58.

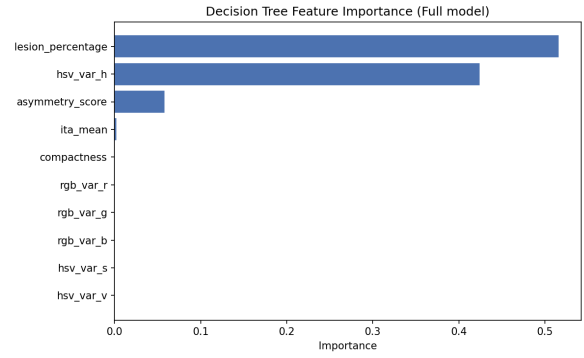


Figure 12: DT Model C Feat Importance

- Adding ITA did not improve the overall performance compared with the model using color variance only.

In terms of feature importance, lesion percentage was the most important feature in all three models. In Model A, lesion percentage had an importance of 0.7292. In Model B, it remained the most important feature with an importance of 0.4293, and in Model C it increased again to 0.5161.

Among the color features, HSV hue variance was the most relevant. It had an importance of 0.3273 in Model B and 0.4240 in Model C. This suggests that variation in hue may help distinguish between benign and malignant lesions.

The ITA mean feature, which represents skin tone, had almost no importance in Model C, with an importance of only 0.0020. This suggests that skin tone did not meaningfully influence the decision tree predictions in this experiment.

The decision tree results suggest that color variance improved the model. Model B achieved the highest test AUC, increasing from 0.647 in the shape-only model to 0.691 when color variance features were included. Even though adding ITA as a skin tone feature did not improve performance based on AUC. It did improve the model in terms of malignant recall, which are the most serious cases, it's also worth noting the high malignant recall of model A considering only shape elements, which was the highest among all 3 models.

6 Conclusion

Can skin colour detection improve the performance of a skin lesion classification model?

After reviewing prior research on dermatological AI bias, we hypothesized that explicitly incorporating skin colour could improve the model's predictive performance. Because a lesion's visual contrast depends heavily on the surrounding skin tone, we assumed that providing the model with colour information—extracted via the Individual Typology Angle (ITA) and superpixel-based HSV/RGB variance—would help the classifier contextualize these differences.

We tested these colour features alongside our baseline geometric metrics, such as asymmetry and com-

pactness. However, we did not observe any significant improvement in classification performance. While skin tone affects visibility, the structural features of the lesion itself seem to carry far more diagnostic weight, meaning the model prioritizes splits on geometric irregularity over colour context.

Furthermore, because the PAD-UFES-20 dataset consists of smartphone images taken under non-uniform lighting conditions, the RGB and HSV variance may have inadvertently captured illumination noise rather than true pathological colour complexity. Consequently, we cannot conclude that basic skin colour detection directly improves the diagnostic model. While skin tone remains a crucial metric for evaluating post-hoc model fairness and bias, simply adding it as a predictive feature does not significantly improve baseline classification.

In a real skin lesion classification problem, the most important mistake to avoid is a false negative. A false negative happens when the model classifies a malignant lesion as benign. This is the most serious error because it could delay diagnosis and treatment. After that, the second most important result is a true positive, because it means the model correctly detects a malignant lesion. False positives are less dangerous, because they classify a benign lesion as malignant, which may lead to unnecessary concern or extra medical exams, but the patient would still be checked. True negatives are useful, but they are the least critical in terms of risk because they only confirm that a benign lesion is benign.

It's important then for us to analyze malignant recalls and not only AUC numbers, the best model in a real world scenario considering this metric would then be the Decision Tree Model A which only uses shape features, because it had the highest malignant recall, detecting 72% of malignant cases. Missing the fewer cancer cases than the other models, is the most important factor in a medical screening context. Although the models that included ITA and colors sometimes improved AUC slightly or kept accuracy close to the other models, they did not improve the detection of malignant lesions, increases in detection of benign lesions were more common. In both Logistic Regression and KNN, adding ITA reduced malignant recall to around 50%, and in the Decision Tree it did not outperform the shape-only model. We can conclude then ITA did not clearly help when the priority is reducing false negatives and detecting as many malignant lesions as possible.

Some future improvements could be made regarding the project, we could look into applying the models into an external dataset with new data, further improvement can be made in regards to masks with multiple lesions, we made a treatment in this project were we only considered the largest lesion, but it would be interesting to maybe separate each lesion and evaluate them individually.

References

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A Appendix

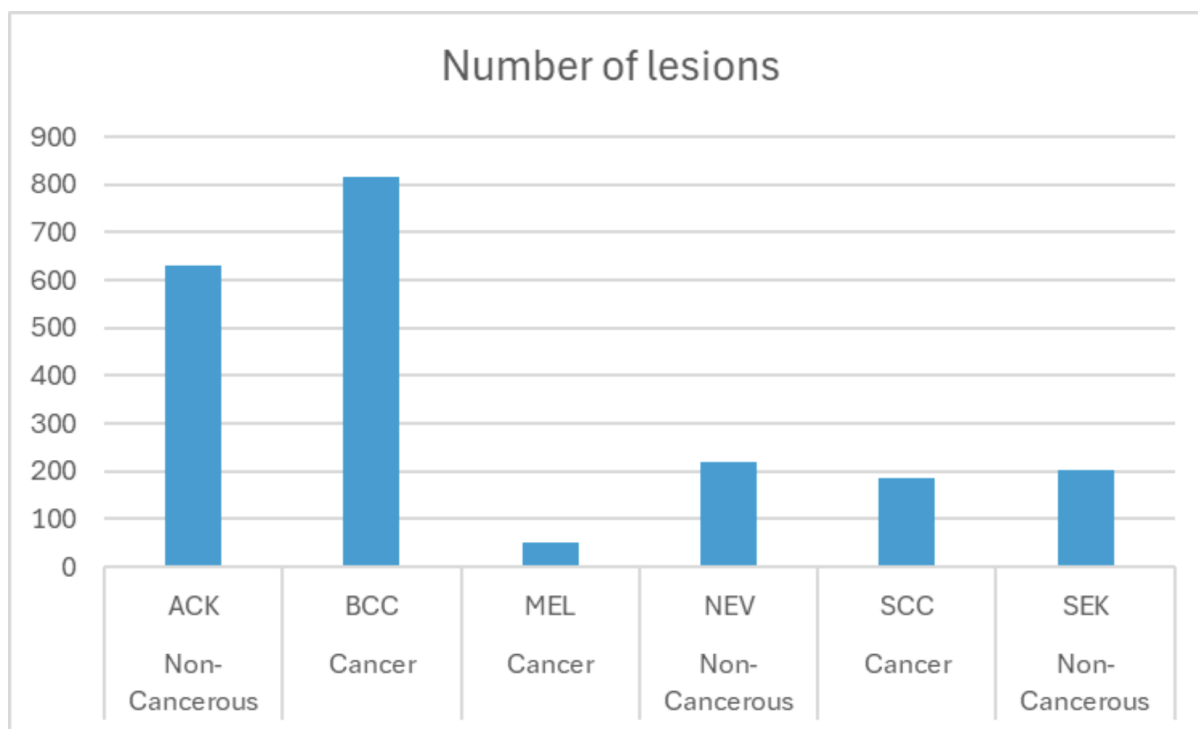


Figure 1: Number of lesions by category

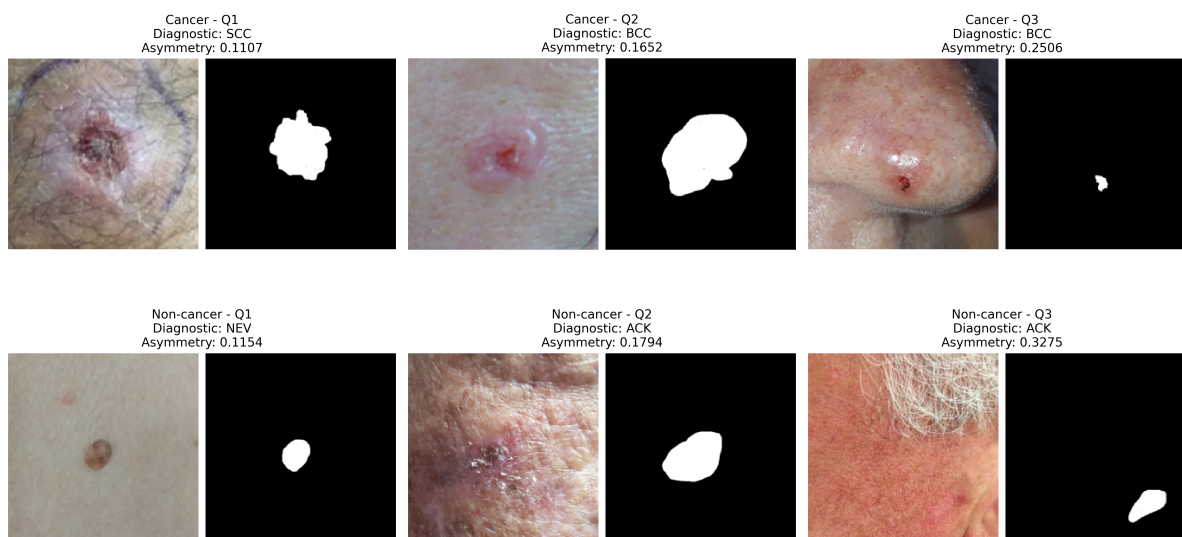


Figure 2: Asymmetry feature extraction examples

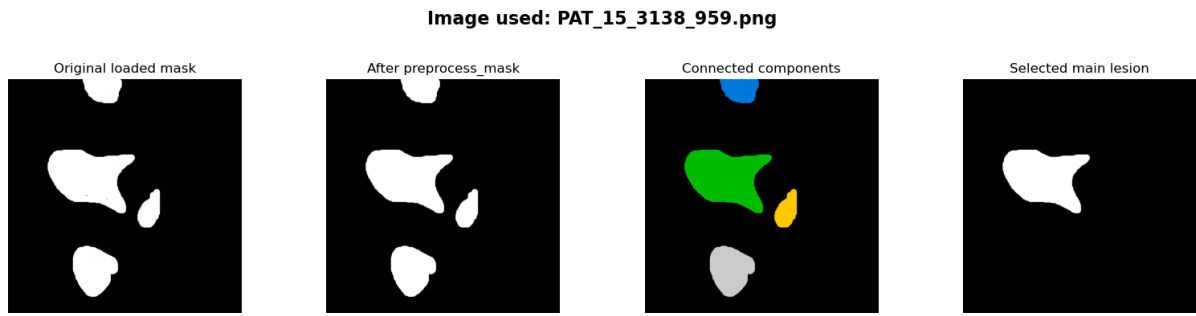


Figure 3: Mask preprocessing to isolate the main lesion

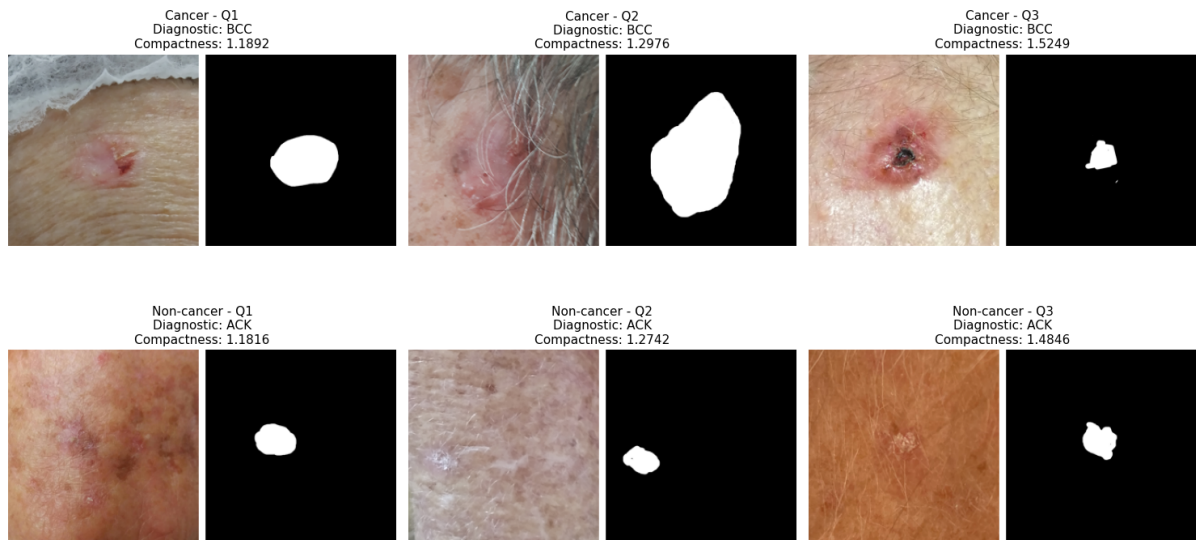


Figure 4: Compactness feature extraction examples

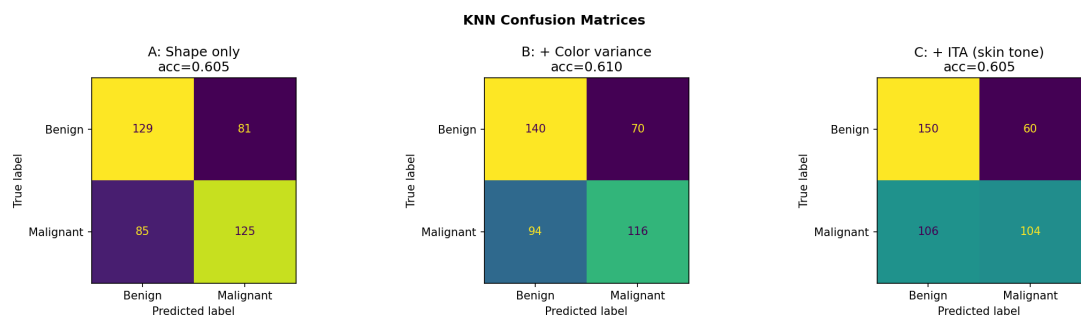


Figure 5: KNN confusion matrices across models

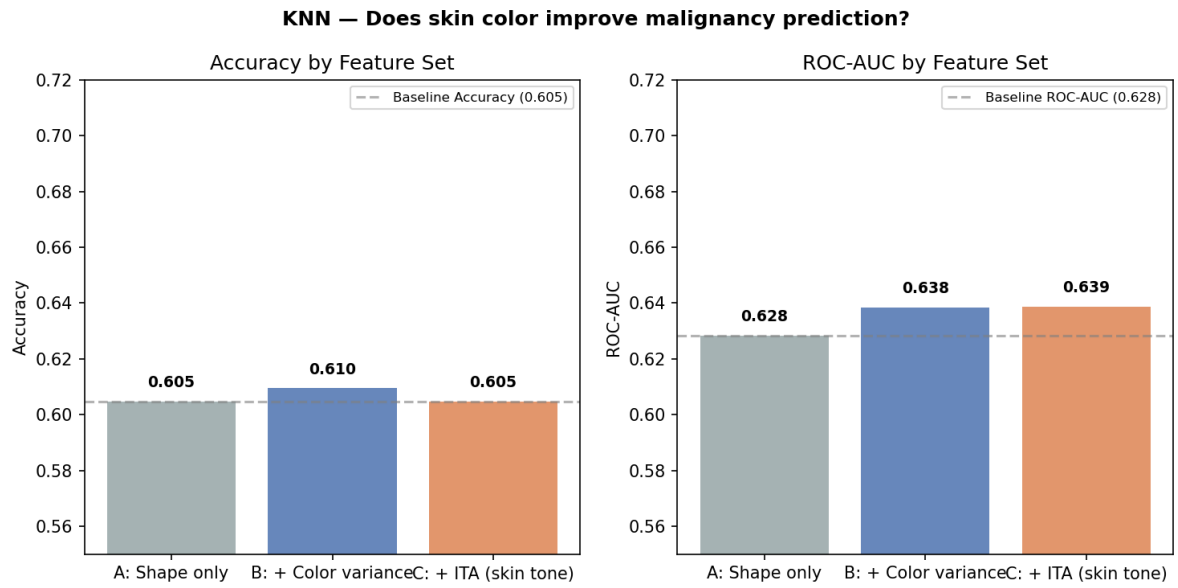


Figure 6: KNN comparison across models

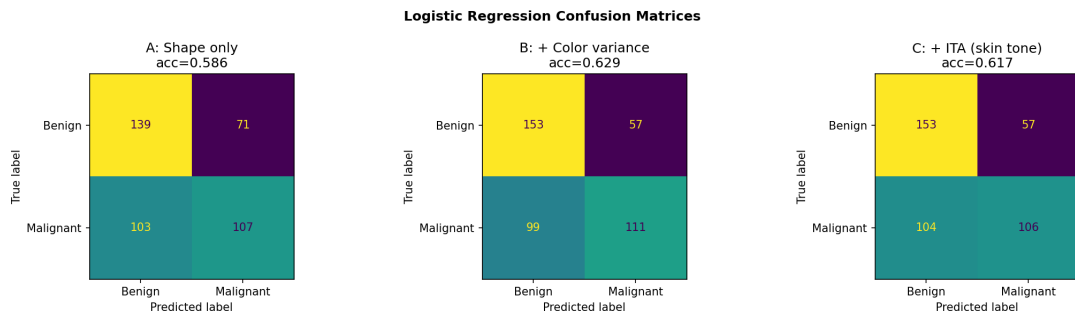


Figure 7: Logistic regression confusion matrices across models

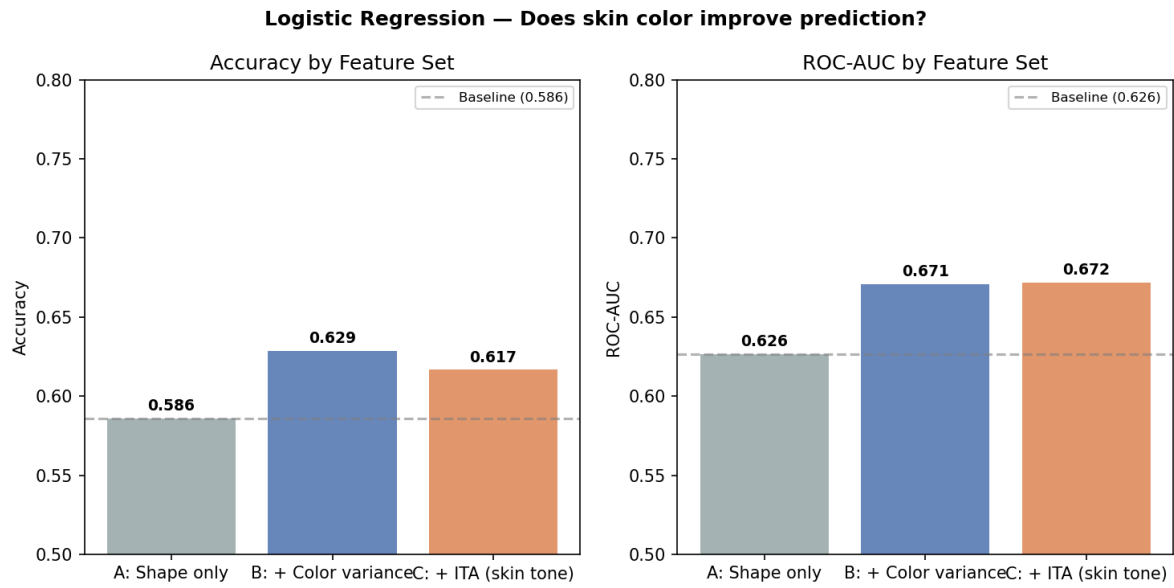


Figure 8: Logistic regression comparison across models

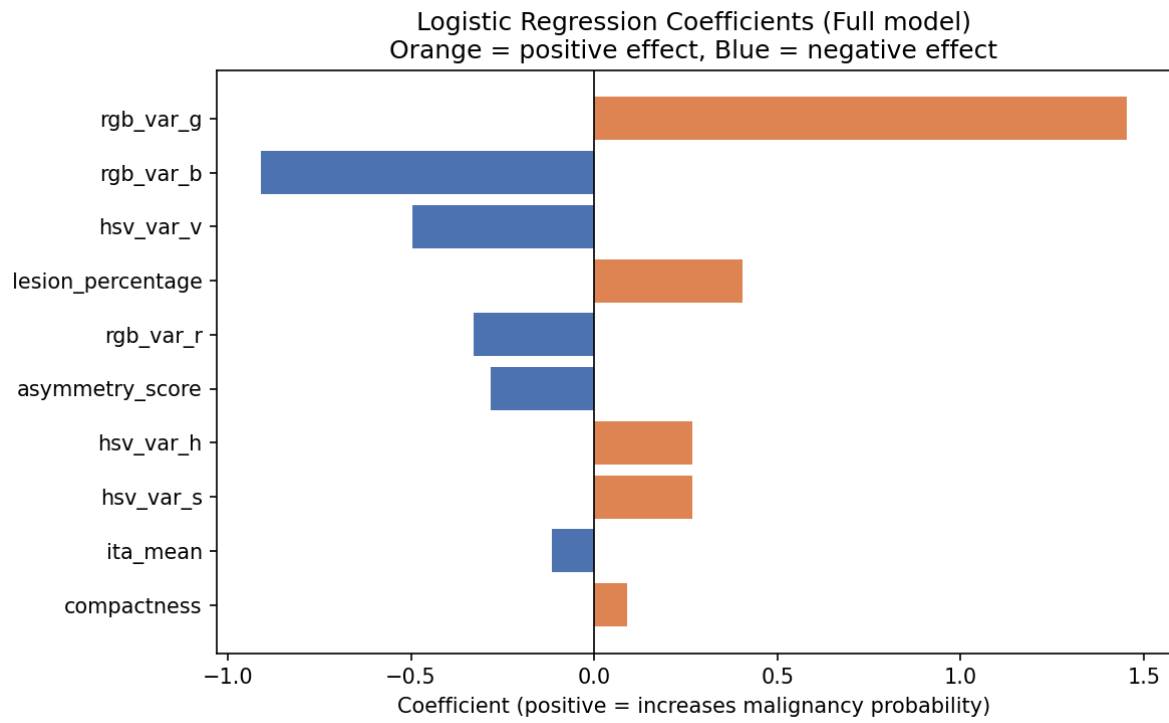


Figure 9: Logistic regression coefficients

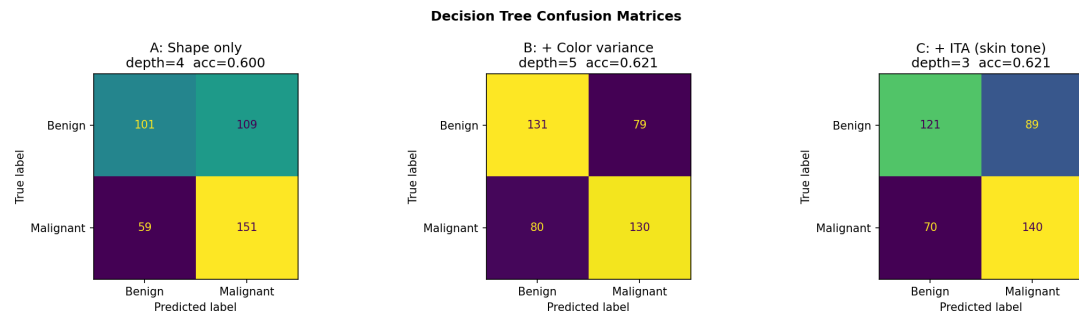


Figure 10: Decision tree confusion matrices across models

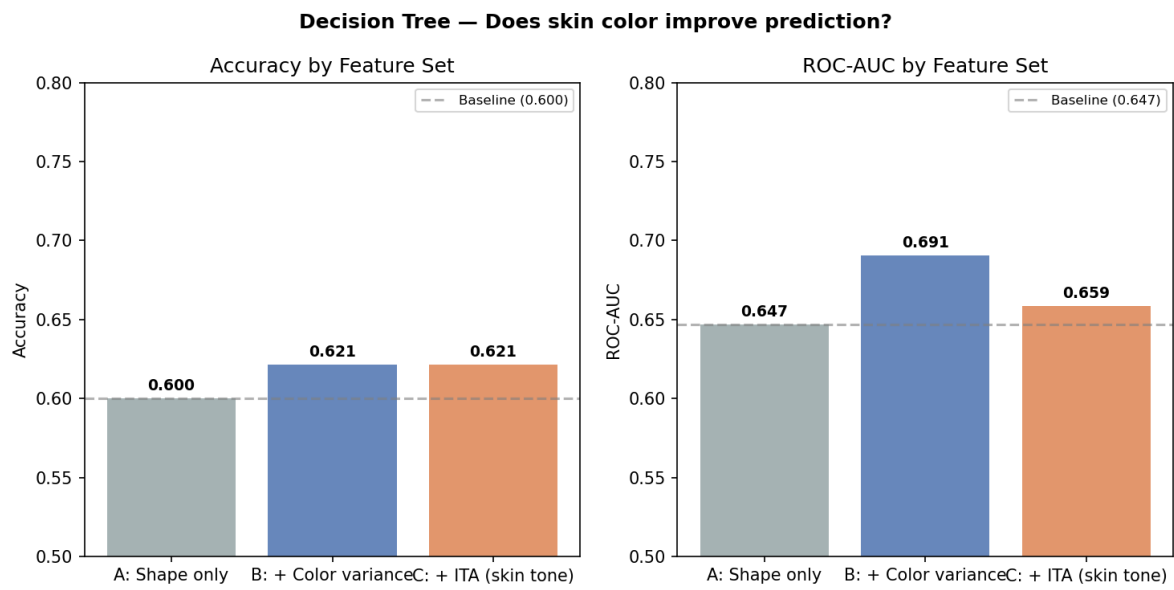


Figure 11: Decision tree comparison across models

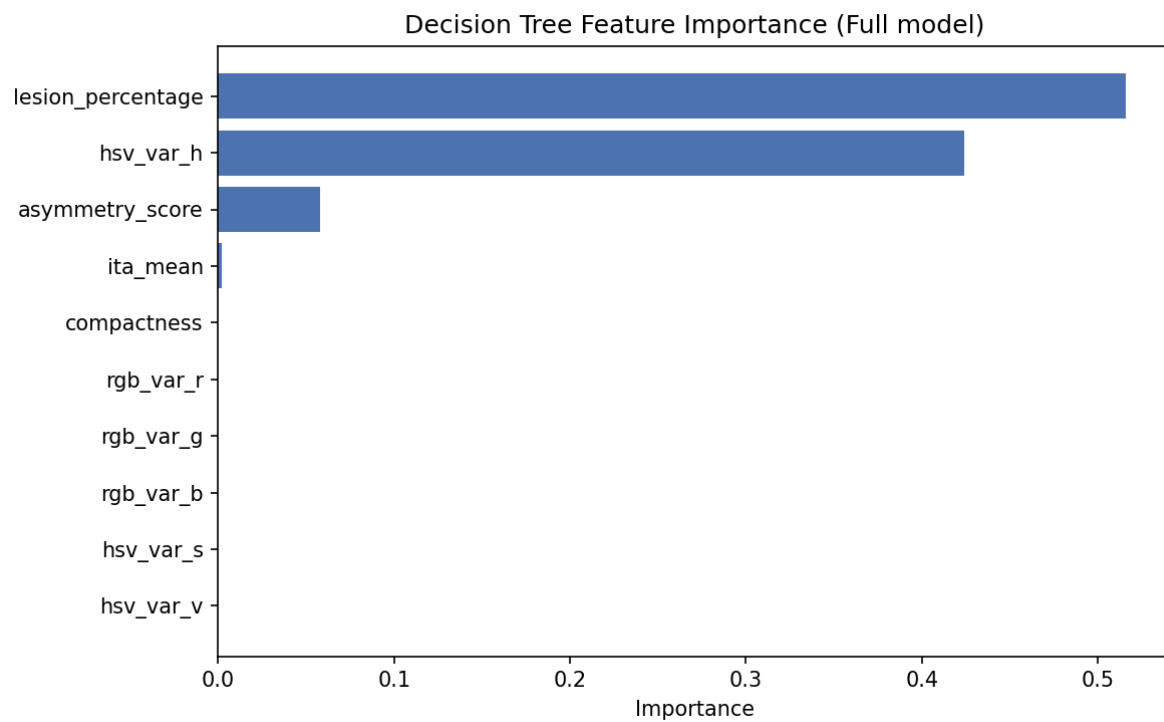


Figure 12: Decision tree Model C feature importance